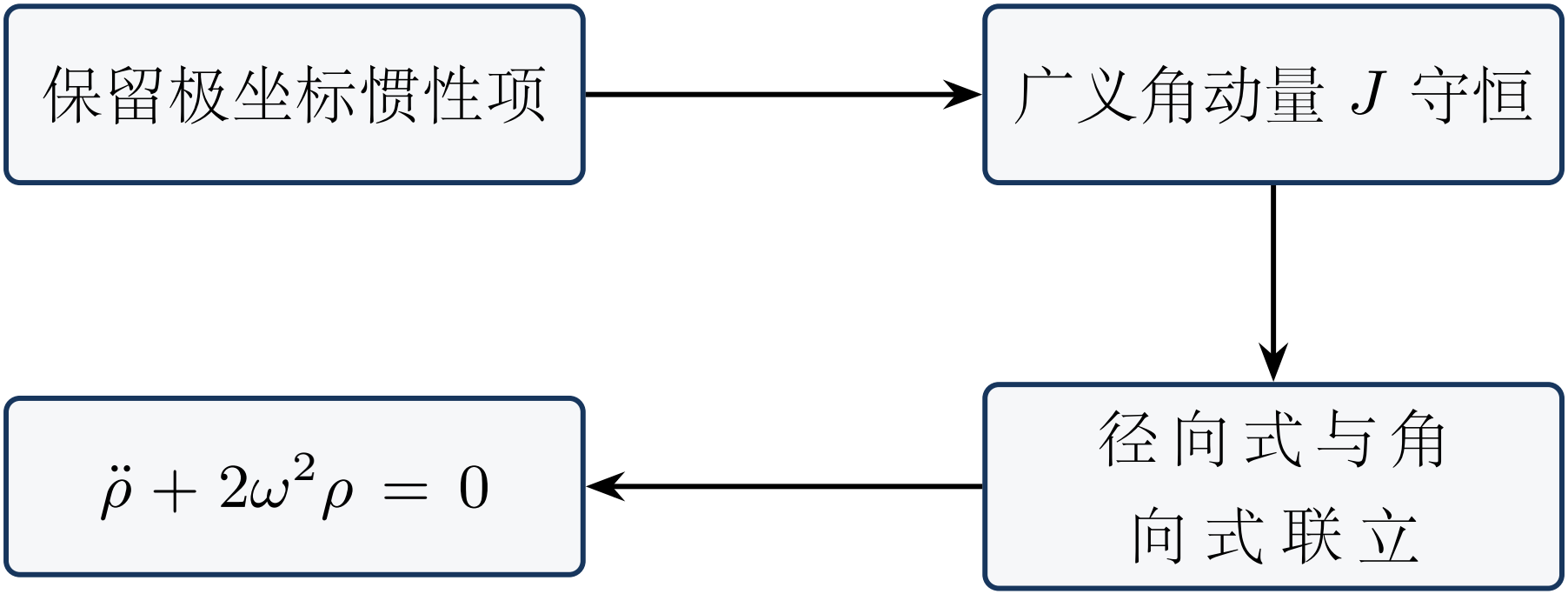

径向振动频率 $\sqrt{2}\omega$；只用瞬时圆轨道条件会漏掉耦合项。